

3.0 Installation: Pipe fittings

George Fischer Signet offers a wide selection of installation fittings that control the position of the magmeter electrodes in relation to the dimensions of the pipe. You will find a complete list of order numbers for installation fittings in the K-factor table in section 7.3.

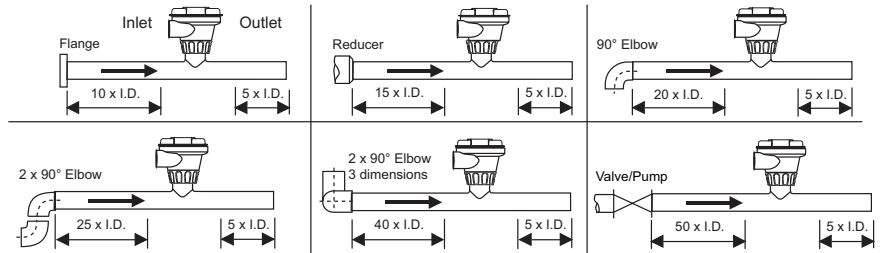
Type	Description	Type	Description
Plastic tees 	0.5 to 4 inch versions - PVC or CPVC	Iron, Carbon Steel, 316 SS Threaded tees 	0.5 to 2 in. versions - Mounts on threaded pipe ends
PVC Glue-on Saddles 	- Available in 10 and 12 inch sizes only - Cut 2-1/2 inch hole in pipe - Weld in place using solvent cement	Carbon steel & stainless steel Weld-on Weldolets 	2 to 4 inch, cut 1-7/16 inch hole in pipe - Over 4 inch, cut 2-1/8 inch hole in pipe
PVC Saddles 	2 to 4 inch, cut 1-7/16 inch hole in pipe 6 to 8 inch, cut 2-1/8 inch hole in pipe	Fiberglass tees & saddles: FPT  FPS 	1.5 in. to 8 in. PVDF insert > 8 in. PVC insert
PP Clamp-on Saddles 	- Available in 10 and 12 inch sizes only - Cut 2-1/8 inch hole in pipe	Metric Union Fitting 	- For pipes from DN 15 to 50 mm - PP or PVDF
Iron Strap-on saddles 	2 to 4 inch, cut 1-7/16 inch hole in pipe - Over 4 inch, cut 2-1/8 inch hole in pipe		

3.1 Location of Fitting

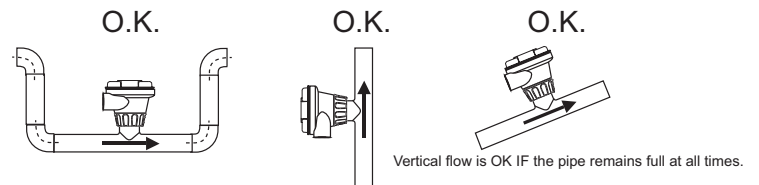
The 2551 can only measure flow if the pipe is full.

If the piping system has entrained air pockets, take steps to locate the sensor so the air pockets will not contact the electrodes.

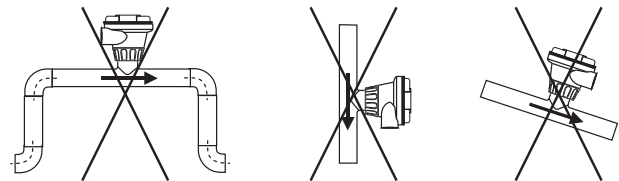
Select a location with sufficient distance of straight pipe immediately upstream of the sensor.



These configurations guarantee that the pipe is always filled. The sensor cannot be exposed to air bubbles.



Avoid these situations unless you are certain that the sensor will not be exposed to air bubbles.



In a gravity-flow system, the tank must be designed so the level does not drop below the outlet. This causes the pipe to draw air in from the tank. If air bubbles pass across the Magmeter electrodes, the output will become erratic.

3.2 Empty Pipe Detection

If the pipe is not full, or the electrodes are not immersed, the flow display will show zero and all output functions will behave accordingly.

If the magmeter electrodes are exposed to flow transitions and air bubbles, the measurement may be incorrect until the conditions are stabilized.

